

Mirror Descent

Recall guarantees for the sub gradient method

$$\begin{aligned} \min; f(x) & : \quad \text{sub-gradient method; } x_{t+1} = x_t - \eta g_t, \quad g_t \in \partial f(x_t) \\ \text{st: } x \in \mathcal{X} & \end{aligned}$$

equivalent: $x_{t+1} = \arg\min: \eta g_t^T x + \frac{1}{2} \|x - x_t\|_2^2$

Thm: If $\eta = \frac{\|x_0 - x^*\|}{L \cdot \sqrt{T}}$, L is Lipschitz constant for f

Then after T steps, sub-optimality:

$$f\left(\frac{1}{T} \sum_{t=1}^T x_t\right) - f(x^*) \leq \frac{\|x_0 - x^*\| \cdot \underbrace{L}_{\text{now we look more closely at } L}}{\sqrt{T}}$$

most of focus of
these lectures

An Example: Robust Regression

$\|Ax - b\|_2^2 \rightarrow$ squared loss is very sensitive to outliers.

↳ Instead: $\min: \|Ax - b\|_1$, also convex.

Need to know: L Lipschitz constant for $f(x) = \|Ax - b\|_1$

$$\left| \|Ax - b\|_1 - \|Ay - b\|_1 \right| \leq L \cdot \|x - y\|_2$$

To simplify calculation: $A = I$, $b = 0$, i.e., $f(x) = \|x\|_1$

$$\left| \|x\|_1 - \|y\|_1 \right| \leq L \cdot \|x - y\|_2$$

$$\text{Take } x = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \mathbf{1}, \quad y = \begin{pmatrix} 1+\epsilon \\ \vdots \\ 1+\epsilon \end{pmatrix} = (1+\epsilon) \cdot \mathbf{1}$$

$$\begin{aligned} \left| n - (1+\epsilon)n \right| &= \epsilon n \leq L \cdot \|x - y\|_2 \\ \epsilon n &\leq L \cdot \epsilon \sqrt{n} \end{aligned}$$

$$\begin{aligned} \|x - y\|_2 &= \|\epsilon \mathbf{1}\|_2 \\ &= \sqrt{\sum \epsilon^2} \\ &= \epsilon \sqrt{n} \end{aligned}$$

This is dimension dependent.

$$\Rightarrow \boxed{L = \sqrt{n}}$$

More generally: hidden dimension dependence

This example shows: $\frac{\|x_1 - x^*\| \cdot L}{\sqrt{n}}$ — conv. guarantee for subgradient method

may hide dimension-dependence

$$\begin{array}{l} \min_x: f(x) \\ \text{s.t. } x \in \mathcal{X} \end{array},$$

suppose it is 1-Lipschitz but with respect to a different norm.

$$\text{Ex: } \underline{\|\nabla f(x)\|_\infty \leq 1} \quad (\text{We have exactly this for } \|\cdot\|_1)$$

Our current bound only work with L as computed using $\|\cdot\|_2$ norm!

$$\|\nabla f(x)\|_\infty \leq 1 \Rightarrow \boxed{\|\nabla f(x)\|_2 \leq \sqrt{n}}$$

Mirror Descent wants to fix this

Idea:

$$x_{t+1} = \operatorname{argmin}_x \eta g_t^T x + \frac{1}{2} \|x - x_t\|_2^2$$

change this.